

Geometric series



- 1 Find the sum of the first 12 terms of the series: $5 + 10 + 20 \dots$
- 2 Find the sum of the first 5 terms of the geometric sequence defined by the following. Round your answers to two decimal places.
 - a $a = 2.187$ and $r = 1.134$
 - b $a = -4.186$ and $r = -2.848$
- 3 Find the sum of the first 11 terms of the series: $1 - 2 + 4 \dots$
- 4 Find the sum of the first 7 terms of the geometric series: $64 + 16 + 4 \dots$
- 5 Find the sum of the first 5 terms of the series: $24 - 12 + 6 \dots$
Round your answer to the nearest whole number.
- 6 Find the sum of the first 7 terms of the series: $8 + 1 + \frac{1}{8} \dots$
Round your answer to three decimal places.
- 7 Consider the series: $5 + \frac{5}{2} + \frac{5}{4} \dots$
 - a Find the common ratio, r .
 - b Find the sum of the first 10 terms, rounding your answer to one decimal place.
- 8 The first 3 terms of a sequence are $x, \frac{2}{5}x^2, \frac{4}{25}x^3$.
 - a Write a simplified expression for the sum of the first n terms of this sequence.
 - b Evaluate the sum when $x = 5$ and $n = 9$.
- 9 Consider the series: $\frac{1}{4} - \frac{1}{5} + \frac{1}{16} - \frac{1}{25} + \frac{1}{64} - \frac{1}{125} + \dots$
Form an expression for the sum of the first $2n$ terms of the series.
- 10 For each of the following series:
 - i Find n , the number of terms in the series.
 - ii Find the sum of the series.



- 11 Find n , the number of terms, in each of the following series:
- a The sum of n terms in the geometric series $2 + 10 + 50 + \dots$ is 195 312.
 - b The sum of n terms in the geometric series $5, -20, 80, \dots$ is 262 145.
 - c The sum of n terms in the geometric series $16 + 4 + 1 + \dots$ is $\frac{1365}{64}$.
 - d The sum of n terms in the geometric series $40, -8, \frac{8}{5}, \dots$ is $\frac{20\,832}{625}$.
- 12 The sum of the first 6 terms of a geometric series is 28 times the sum of its first 3 terms. Find r , the common ratio.
- 13 The sum of the first 9 terms of a geometric series is 436 905, and the common ratio is $\frac{1}{4}$. Find a , the first term in the series.
- 14 The first term of a geometric sequence is 9 and the fourth term is 243. What is the sum of the first 13 terms of the corresponding series?
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Infinite series

- 15 What condition must be satisfied by an infinite geometric series in order for its sum to exist?
- 16 For each of the following infinite geometric sequences:
- i Find the common ratio.
 - ii Determine whether the sum is divergent or convergent.
- | | |
|--------------------------|-------------------------|
| a 11, 22, 44, 88, ... | b 3, -12, 48, -192, ... |
| c -40, -20, -10, -5, ... | d 256, 64, 16, 4, ... |
- 17 For each of the following infinite geometric sequences:
- | | |
|--|---|
| i Find the common ratio, r . | ii Find the limiting sum of the series. |
| a $2, \frac{1}{2}, \frac{1}{8}, \frac{1}{32}, \dots$ | b 125, 25, 5, 1, ... |
| c 16, -8, 4, -2, ... | d $28, -7, \frac{7}{4}, -\frac{7}{16}, \dots$ |



18 For a particular geometric sequence, $a = 2$ and $a_{\infty} = 4$.

a Find r , the common ratio.

b Find the first 3 terms in the sequence.

19 Consider the infinite sequence: $1, x - 7, (x - 7)^2, (x - 7)^3, \dots$

Determine the range of values of x if the infinite series has a limiting sum.

20 The limiting sum of the infinite sequence $1, 4x, 16x^2, \dots$ is 5. Find the value of x .

21 Consider a geometric series in which all terms are positive that has first term a and common ratio r . The sum of the infinite series is 768. The sum of the first 4 terms is 765.

Find the first three terms of the sequence.

22 Consider the infinite geometric series: $6 + 2 + \frac{2}{3} + \frac{2}{9} + \dots$

a Find the number of terms, n , that would be required to give a sum of $\frac{177\,146}{19\,683}$.

b Find the infinite sum of the series.

23 Consider the infinite geometric series: $5 + \sqrt{5} + 1 + \dots$

a Find the common ratio, r , expressing your answer with a rational denominator.

b Find the limiting sum, expressing your answer with a rational denominator.

24 Find the limiting sum of the infinite series $\frac{1}{5} + \frac{3}{5^2} + \frac{1}{5^3} + \frac{3}{5^4} + \frac{1}{5^5} + \dots$

Sigma notation

25 Evaluate:

a

$$\sum_{i=1}^4 4^i$$

b

$$\sum_{j=1}^5 243 \left(\frac{5}{3}\right)^j$$

c

$$\sum_{j=1}^5 32 \left(\frac{1}{2}\right)^j$$

d

$$\sum_{i=3}^8 (-3)^i$$

e

$$\sum_{i=2}^7 (-2^i)$$

f

$$\sum_{i=1}^5 (2(3)^{i-1})$$

g

$$\sum_{j=4}^8 \left(\frac{1}{4}(3)^{j-1}\right)$$

h

$$\sum_{j=1}^4 \left(-3 \left(\frac{1}{3}\right)^j\right)$$

26 Write the series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{1}{1024}$ in sigma notation.



27 Find the value of the following:

a

$$\sum_{i=1}^{\infty} 3 \left(\frac{1}{4} \right)^{i-1}$$

b

$$\sum_{i=1}^{\infty} 5 \left(-\frac{1}{4} \right)^{i-1}$$

c

$$\sum_{i=1}^{\infty} \frac{1}{8} \left(-\frac{1}{2} \right)^{i-1}$$

d

$$\sum_{i=1}^{\infty} -\frac{1}{2} \left(\frac{5}{7} \right)^{i-1}$$

e

$$\sum_{i=1}^{\infty} (0.9)^i$$

f

$$\sum_{k=1}^{\infty} 9^{-k}$$

Repeating decimals

28 The repeating decimal $0.8888\dots$ can be expressed as a fraction when viewed as an infinite geometric series.

- a Using fractions, write the first five terms of the geometric sequence representing $0.8888\dots$
- b State the values of the first term a and the common ratio r of this sequence.
- c Calculate the infinite sum of the sequence as a fraction.

29 The repeating decimal $0.444444\dots$ can be expressed as a fraction when viewed as an infinite geometric series.

- a Express the first two decimal places, 0.44 , as an unsimplified fraction.
- b Express the second two decimal places, 0.0044 , as an unsimplified fraction.
- c Express the third two decimal places, 0.000044 , as an unsimplified fraction.
- d Now, state the values of the first term a and the common ratio r of the sequence formed from these first three terms.
- e Calculate the infinite sum of the sequence as a fraction.

30 For each of the following repeating decimals:

- i Rewrite the decimal as a geometric series.
- ii Express the decimal as a fraction.

a $0.6666\dots$

b $0.06666\dots$

31 Consider the sum $0.25 + 0.0025 + 0.000025$



a Express the sum as a decimal.

b Now, express the sum as a fraction.

32 Consider the repeating decimal $0.2222\ldots$ in the form

$$\frac{2}{10} + \frac{2}{100} + \frac{2}{1000} + \frac{2}{10\,000} + \ldots$$

Rewrite the repeating decimal as a fraction.

33 Consider the repeating decimal $0.575\,75\ldots$ in the form

$$0.57 + 0.0057 + \ldots$$

Rewrite the repeating decimal as a fraction.

34 Find the value of $0.994 + 0.000\,994 + 0.000\,000\,994\ldots$

Applications

35 At the start of 2014, Pauline deposits \$5000 into an investment account. At the end of each quarter, she makes an extra deposit of \$700.

The table below shows the first few quarters of 2014. All values in the table are in dollars:

Quarter	Opening Balance	Interest	Deposit	Closing Balance
Jan - Mar	5000	200	700	5900
Apr - Jun	5900	236.00	700	6836.00
Jul - Sep	6836.00	273.44	700	7809.44

a Find the quarterly interest rate.

b Write an expression for the amount in the account at the end of the first quarter.

c Write an expression for the amount in the account at the end of the second quarter.

d Write a similar expression for the amount in the account at the end of the third quarter.

e Write an expression for the amount in the investment account after n quarters.

f Now, determine the total amount in Pauline's account at the beginning of 2016 to the nearest dollar.



- 39** Rochelle invests \$190 000 at a rate of 7% per annum compounded annually, and wants to work out how much she can withdraw each year to ensure the investment lasts 20 years.
- a** Write an expression for A_1 , the balance of the account after 1 year. Use x to represent the amount to be withdrawn each year.
 - b** Write an expression for:
 - i** A_2
 - ii** A_3
 - c** Write an expression for A_n .
 - d** Now, determine Rochelle's annual withdrawal amount.
- 40** Mario invests \$110 000 at a rate of 16% per annum compounded quarterly and wants the investment to last 25 years.
- a** The amount in the account after n quarters can be expressed as the n th term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n quarters. Use x to represent the amount to be withdrawn each quarter.
 - b** Now, determine Mario's quarterly withdrawal amount.
- 41** Emma invests \$190 000 at a rate of 3% per annum compounded monthly, and wants to work out how much she can withdraw each month to ensure the investment lasts 30 years.
- a** The amount in the account after n months can be expressed as the n th term of a geometric sequence minus the sum of a different geometric sequence.
Write an expression for the amount in the investment account after n months. Use x to represent the amount to be withdrawn each month.
 - b** Now, determine Emma's monthly withdrawal amount.